

**Jharkhand University of Technology, Ranchi**  
**Diploma 3rd Semester Examination, 2024**

**Subject : Engineering Mathematics-III**

**Subject Code : 301**

**Time Allowed : 3 Hours**

**Full Marks : 80**

*Answer in your own words.*

*Answer five questions in which Question No. 1 is compulsory.*

*The figure in the margin indicates full marks.*

*All questions carry equal marks.*

**1. Choose the correct option/answer the following:**

**2×8=16**

(i) Find the value of the integral  $\int_{\pi/4}^{\pi/2} \cot x dx$ .

(a)  $\log 2$

(b)  $2 \log 2$

~~(c)  $\frac{1}{2} \log 2$~~

(d) None of these

(ii) The integrating factor of the different equation  $\frac{dx}{dy} + \frac{3x}{y} = \frac{1}{y^2}$  is

(a)  $x^3$

(b)  $-y^3$

~~(c)  $y^3$~~

(d) None of these

(iii) The number of strips required in Simpson's 1/3 rd rule is a multiple of

(a) 1

(b) 3

~~(c) 2~~

(d) 6

(iv) As soon as a new value of a variable is found by iteration, it is used immediately in the following equations, this method is called

(a) Gauss Jordan method.

(b) Gauss Jacobi method.

~~(c) Gauss Seidel method.~~

(d) None of these

(v) The variance of the binomial distribution with n observations and probability of success p is

(a) np

(b)  $\sqrt{npq}$

~~(c) npq~~

(d) None of these

(vi) The Laplace transform of  $t^3$  is

~~(a)  $\frac{6}{s^4}$~~

(b)  $\frac{24}{s^4}$

(c)  $\frac{3}{s^3}$

(d) None of these

(vii) A set of values of the variables satisfying all the constraints in an LPP is known as

- (a) Objective function. ~~(b) Feasible solution.~~  
(c) Linear solution. (d) No solution.

(viii) If  $f(x)$  is an odd function then the Fourier series expansion of  $f(x)$  will contain

- ~~(a) only sine terms~~ (b) Both sine and cosine terms.  
(c) only cosine terms (d) None of these.

2. (a) Evaluate the integral :  $\int \frac{2x}{(2x+1)^2} dx$ . 8

(b) Find the area of the region bounded by the curve  $y = x^2$ , the  $x$ -axis and the lines  $x = 1$  and  $x = 3$ .  $\frac{26}{3}$

3. (a) Solve  $\frac{dy}{dx} = \sin(x+y) + \cos(x+y)$ . 8

(b) If the temperature of the air is  $30^\circ\text{C}$  and the substance cools from  $100^\circ\text{C}$  to  $70^\circ\text{C}$  in 15 minutes. Find when the temperature will be  $40^\circ\text{C}$ . 8+8

4. (a) Find the root of the equation  $\cos x = xe^x$  using the regular falsi method correct to four decimal places.

(b) From the following table of half yearly premium for policies maturing at different ages. Using Newton's forward interpolation formula estimate the premium for policies maturing at the age of 46.

| Age         | 45     | 50    | 55    | 60    | 65    |
|-------------|--------|-------|-------|-------|-------|
| Premium Rs. | 114.84 | 96.16 | 83.32 | 74.48 | 68.48 |

5. (a) A can hit a target 3 times in 5 shots, B 2 times in 5 shots and C 3 times in 4 shots. They fire a volley. What is the probability that (i) two shots hit (ii) at least two shots hit? 8+8

(b) The incidence of occupational disease in an industry is such that the workmen have a 10% chance of suffering from it. What is the probability that in a group of 7, five or more will suffer from it? 8+8

6. (a) Find the inverse Laplace transform of  $\frac{3s+2}{s^2-s-2}$ . 8

(b) Solve the following problem graphically:

Minimize and maximize  $z = 3x + 9y$

Subject to the constraints  $x + 3y \leq 60, x + y \geq 10, x \leq y, x, y \geq 0$ . 8+8

8. Write Short notes on any four of the following: 4x4=16

- (a) Integration by parts  
(b) Linear differential equation  
(c) Newton-Raphson's method  
(d) Addition Theorem of Probability  
(e) Fourier Euler's formula